مواصفات تطبيق إنجاز | Enjaz — Full App Specification

Prepared for Mohamed Gamal — Version 1.0 — Target: Android, built by an AI coding agent (Google Antigravity)

> ملاحظة بالعربي: الملف ده مكتوب بالإنجليزي عن قصد عشان الوكيل (Antigravity) يفهمه بدقة، لكن كل نصوص الواجهة اللي هتظهر في التطبيق مكتوبة جواه بالعربي والإنجليزي. لو أي جزء مش واضح، اسألني في الشات.

0. Instructions to the AI coding agent (read first)

- The owner is NOT a programmer. Talk to him in simple Egyptian Arabic, no jargon. Explain every step he must do (run, install, test) as short numbered steps.
- Build in the phases listed in section 11. Finish one phase, give a plain-Arabic test checklist, wait for his approval, commit to Git, then continue.
- Do not add features that are not in this spec. If something is ambiguous, or you want to change a decision, ask first.
- Follow the coding rules in section 10.
- The UI mockup image supplied with this document is the visual reference. Match its layout, colors and spacing. Where the image and this document conflict on behavior, this document wins.

1. Product overview

Working name: إنجاز / Enjaz (changeable). A fully offline, single-user Android app that replaces the owner's phone notes for tracking his work and life.

Problems it solves:

1. Notes do not record when a task was done, when it started, or when it finished.
2. Notes do not record partial progress or why something stopped.
3. There is no place for waiting or postponed items and the reason (for example: no money yet).
4. Long-term goals that are completed in parts have no progress tracking.
5. There is no dashboard answering: what did I achieve this week / month / year?
6. Ideas to execute later are mixed in with tasks.
7. Daily tasks and errands need very fast logging.

Non-goals for v1: no login, no cloud sync, no ads, no internet permission, no social features, no AI features inside the app, no Google Play release (the owner installs the APK directly).

2. Tech stack

- Flutter (latest stable), Dart, Material 3, Android only (minSdk 24, target = latest).
- Local database: SQLite through Drift, with versioned migrations.
- State management: Riverpod. Navigation: go_router.
- Localization: flutter_localizations + ARB files (ar, en). Default follows the device language; a setting can force a language.
- Charts: fl_chart. Notifications: flutter_local_notifications + timezone. Backup: share_plus, file_picker, path_provider. App lock: local_auth (phase 7).
- Fonts bundled as assets (never downloaded at runtime): Cairo for Arabic, Inter for English.
- Folder structure: feature-first. lib/features/{today, tasks, goals, ideas, stats, settings} and lib/core/{db, theme, l10n, utils}.
- Time: store timestamps as UTC epoch milliseconds. Store due dates as local date strings YYYY-MM-DD and times as HH:mm. The day boundary is local midnight.

3. Design system (minimal style)

Principles: clean, lots of white space, one accent color, rounded cards, thin outlined icons (Material Symbols Rounded), calm motion (150–250 ms), minimum touch target 48 dp, usable at 1.3x font scale.

RTL: Arabic is primary, so the whole layout mirrors in Arabic (RTL) and stays LTR in English. Bilingual labels: in Arabic mode show the Arabic label with a small English subtitle only where space allows (bottom navigation, section headers, empty states). In English mode show English only.

Token	Light	Dark
background	#FAFAF9	#0F1412
surface (cards)	#FFFFFF	#1A1F1E
surface-variant	#F5F5F4	#232928
text	#1C1917	#F5F5F4
text-muted	#78716C	#A8A29E

Token	Light	Dark
border	#E7E5E4	#2F3534
accent (teal)	#0D9488	#2DD4BF

Status colors (light / dark): todo #78716C / #A8A29E; in_progress #2563EB / #60A5FA; partial #D97706 / #FBBF24; done #16A34A / #4ADE80; postponed #6B7280 / #9CA3AF; waiting #7C3AED / #A78BFA; stopped #DC2626 / #F87171.

Shape and type: card radius 16, chips fully rounded, bottom sheets 24 radius on top, spacing in multiples of 4. Use a 1 px border instead of shadows in light mode and no shadows in dark mode. Type scale: title 22 semi-bold, section header 16 semi-bold, body 14–15, caption 12.

Theme setting: Light / Dark / Follow system. Both themes are first-class and must be built from the start (no hardcoded colors anywhere).

Empty states: a simple icon, one friendly line, and one action button.

4. Data model (Drift tables)

Every table has: id (int primary key, auto-increment), created_at, updated_at (epoch ms), deleted_at (nullable, soft delete; trash is purged after 30 days).

tasks

- title (text, required), notes (text, nullable)
- kind: task | errand
- status: todo | in_progress | partial | done | postponed | waiting | stopped (default todo)
- priority: low | normal | high (default normal)
- due_date (nullable; null means the task is in the no-date Inbox), due_time (nullable)
- started_at, finished_at (nullable)
- progress (int 0–100, default 0)
- reason_id (FK to reasons, nullable), reason_text (nullable) — the current reason for postponed / waiting / stopped / partial
- outcome_note (nullable) — how it ended, free feedback
- postponed_until (nullable date), postpone_count (int, default 0)
- location_text (nullable, used for errands)
- goal_id, milestone_id (FK, nullable)
- reminder_offset_minutes (nullable), recurrence_id (FK, nullable)

task_events (append-only history log)

- task_id, at (epoch ms), from_status, to_status, progress, reason_id, reason_text, note
- Every status change writes one event. The dashboard is computed from events and tasks.

reasons

- label_ar, label_en (nullable), category: postpone | waiting | stopped | partial | any, is_default, usage_count, archived
- Seed data (AR / EN): مفيش فلوس / No money; مستني حد / Waiting for someone; مفيش وقت / No time; مفيش طاقة أو مود / No energy; ظرف طارئ / Emergency; محتاج معدات / Need equipment; مستني رد العميل / Waiting for client; الطقس / Weather; ظهرت أولوية أهم / Higher priority came up; محتاج معلومات / Need information.
- The user can add, edit and archive reasons. A reason used by any event is archived, never deleted.

goals

- title, description, category (free text)
- progress_mode: milestones | numeric
- target_value, current_value, unit (numeric mode only; example: 5000 / جنيه)
- start_date, target_date (nullable)
- status: active | paused | done | abandoned, completed_at, status_reason (for paused / abandoned)

milestones: goal_id, title, notes, sort_order, weight (int, default 1), due_date (nullable), status (pending | done), completed_at.

goal_updates: goal_id, at, note, value_delta (nullable, numeric mode progress log).

Goal progress: milestones mode = (sum of weights of done milestones) / (sum of all weights) x 100. Numeric mode = current_value / target_value x 100, capped at 100.

ideas: title, description, status: new | exploring | converted | archived, converted_type (task | goal, nullable), converted_id (nullable). **tags** (id, name) and **idea_tags** (idea_id, tag_id).

recurrences: rule (daily | weekly | monthly), weekdays (bitmask), day_of_month, starts_on, ends_on (nullable), template fields (title, kind, due_time, notes, priority, reminder_offset_minutes), last_generated_until.

settings: key / value (language, theme, week_start, numerals, daily_summary_enabled and time, evening_review_enabled and time, lock_enabled, backup_reminder).

Index these columns: tasks.status, tasks.due_date, tasks.deleted_at, tasks.goal_id, task_events.at, task_events.task_id.

5. Status system and rules

Key	Arabic	English	Icon	Required when entering
todo	لسه	To do	circle	nothing
in_progress	جارية	In progress	play_circle	sets started_at if empty
partial	اتعمل جزء منها	Partially done	pie_chart	progress 1–99 AND a reason for not finishing; note optional
done	تمت	Done	check_circle	sets finished_at = now and progress = 100; outcome note optional
postponed	مؤجلة	Postponed	schedule	reason required; postponed_until optional (default tomorrow); postpone_count + 1
waiting	منتظرة	Waiting	hourglass_empty	reason required (what am I waiting for); review date optional
stopped	متوقفة	Stopped	stop_circle	reason required

Transition rules:

1. Any status can change to any other status (no locked flow). Every change writes a task_event.
2. Moving to done sets finished_at to now (editable afterwards). Duration is shown only when both started_at and finished_at exist.
3. Reopening a done task (to todo or in_progress) clears finished_at. The old events stay in the history.
4. Moving from postponed to any other status clears postponed_until.
5. A postponed task whose postponed_until is today or earlier appears in Today under the section 'مؤجلة وحان موعدها'. A task is overdue when due_date is before today and the status is todo, in_progress or partial.
6. Quick checkbox on a card sets done immediately with no dialog. A snackbar shows 'تراجع / Undo' (8 seconds) and 'إضافة ملاحظة / Add note'.
7. Undo after any status change restores the task and removes the event it created.

6. Screens

Global navigation: bottom bar with 4 tabs: اليوم / Today, الأهداف / Goals, الأفكار / Ideas, الإحصائيات / Stats. A '+' floating button on Today, Goals and Ideas adds an item of that tab's type. Stats has no button. Top bar on Today has two icons: History (السجل) and Settings (الإعدادات).

6.1 Today (اليوم) — home

- Header: date switcher (previous / next arrows, tap to open a date picker, default today) showing the Arabic weekday and full date. A progress ring shows done / total for that day, for example '٣ من ٨ تمت / 3 of 8 done'.
- A small chip 'بدون تاريخ / No date' with a count opens the Inbox list of undated tasks.
- Sections in this order (hide empty ones):
  1. متأخرة / Overdue — unfinished items from earlier days, with a one-tap 'نقل لليوم / Move to today'.
  2. مهام اليوم / Tasks — kind = task, due on the selected day.
  3. مشاوير اليوم / Errands — kind = errand, with a pin icon and the location text.
  4. مؤجلة وحان موعدها / Postponed and due now.
  5. منتظرة ومؤجلة / Waiting and postponed — collapsed, count badge; lists every waiting item (any date) and postponed items with a future date.
  6. تمت اليوم / Done today — collapsed.
- Task card: leading checkbox (tap = done, long-press = status sheet), title (one line), time, priority dot, status chip, a thin progress bar for in_progress / partial, one line with the reason for postponed / waiting / stopped / partial, a small goal tag if linked.
- Swipe: toward the start edge = Done, toward the end edge = Postpone (opens the status sheet on postponed). Swipe directions must mirror correctly in RTL.
- Tap = Task details. Long-press = status sheet.
- Empty state: 'مفيش حاجة مخططة النهارده / Nothing planned today' with the button 'أضف مهمة / Add task'.

6.2 Quick add sheet (إضافة سريعة)

- Bottom sheet from the '+' button. Title field is auto-focused. A segmented control: مهمة / Task, مشوار / Errand, هدف / Goal, فكرة / Idea. Choosing Goal or Idea switches to that item's short form (title, optional description, optional target date for goals).
- For task and errand: date chips (اليوم / Today, بكرة / Tomorrow, اختر تاريخ / Pick date, بدون تاريخ / No date), optional time, priority, location (errand only), optional goal link, optional reminder.

- Buttons: 'حفظ / Save' and 'حفظ وإضافة آخر / Save and add another'. The keyboard action key also saves.
- Multi-line paste: if the title contains several lines, offer 'تحويلها إلى N مهام؟ / Create N tasks?' (this helps move the owner's old phone notes into the app).

6.3 Status sheet (تغيير الحالة)

- Opened by long-press or from Task details. Shows the 7 statuses as a grid. After a status is picked, the fields required by section 5 appear inside the same sheet: reason quick-chips (from the reasons table, filtered by category, plus 'سبب آخر / Other' for free text), a progress slider for partial, a date picker for postponed, and an optional note.
- 'تأكيد / Confirm' stays disabled until the required fields are valid.

6.4 Task details (تفاصيل المهمة)

Top to bottom: editable title and kind toggle; status chip (tap opens the status sheet); Timeline card with Created, Started, Finished (each editable) and a computed Duration; progress slider (visible for in_progress and partial); Reason card (visible for postponed, waiting, stopped, partial) showing the reason and since when; Outcome note ('إزاي خلصت؟ / How did it end?', always visible once done); Schedule (date, time, reminder, repeat); priority; location; link to goal and milestone; notes; History list (every event with time, reason and note); actions: duplicate and delete (soft delete with undo).

6.5 History (السجل) — all tasks

- Search box (searches title, notes and reasons). Filter chips: status (multi-select), kind, date range, goal, has reason. Sort: newest or oldest. Items grouped by day. Tap opens Task details. Purpose: answer 'when did I do X and what happened'.

6.6 Goals (الأهداف)

- Three segments: نشطة / Active, متوقفة / Paused, مكتملة / Done.
- Goal card: title, progress bar with percentage, '4 من ١٠ مراحل / 4 of 10 milestones' or '١٢٠٠ من ٥٠٠٠ جنيه' for numeric goals, target date with days left (red when overdue), and the name of the next pending milestone.

6.7 Goal details

- Big progress ring. Editable title, description, dates and progress mode.
- Milestones list: drag to reorder, tick to complete (stores completed_at), add / edit / delete, optional weight in an advanced field.

- Numeric mode: button 'تسجيل تقدم / Log progress' (value and note) and a small line chart of progress over time.
- Linked tasks list with their statuses and the button 'إضافة مهمة لهذا الهدف / Add task to this goal'.
- Activity log (milestones completed, progress updates, status changes).
- Actions: pause (reason required), resume, mark done, abandon (reason required). When the last milestone is ticked, ask 'الهدف خلص؟ / Mark goal as done?'.

6.8 Ideas (الأفكار)

- Search, tag filter chips and a status filter: جديدة / New, بتتدرس / Exploring, اتحولت / Converted, مؤرشفة / Archived. Two-column masonry cards: title, three-line description preview, tags, date.
- Idea details: edit title, description, tags and status. Buttons 'حوّل لهدف / Convert to goal' and 'حوّل لمهمة / Convert to task' open the quick-add form pre-filled. After saving, the idea becomes converted and links to the created item (and the item links back). Archive and delete are available.

6.9 Stats (الإحصائيات)

See section 7.

6.10 Settings (الإعدادات)

Language; Theme; Week start (default Saturday); Numerals (Western 123 or Arabic-Indic ١٢٣, default Western); Notifications (daily summary on/off and time, evening review on/off and time); Reasons manager; Tags manager; Backup and export; App lock; Trash (restore or purge deleted items); About (version).

7. Stats dashboard (الإحصائيات)

Range control: a segmented control أسبوع / Week | شهر / Month | سنة / Year, with previous / next arrows and a label such as '٢٥ أكتوبر – ١ نوفمبر'. The week starts on the day set in Settings (default Saturday).

Definitions (implement exactly, in a pure Dart service with unit tests):

- **Scheduled set** = non-deleted tasks whose due_date falls inside the range. Tasks with no due_date are counted by their created date.
- **Completed** = tasks that have a 'done' event whose timestamp falls inside the range (counted by finished date, regardless of due date).

- **Completion rate** = (scheduled-set tasks that are done) / (scheduled-set size), excluding scheduled tasks whose date is still in the future.
- **Postponed count** = number of distinct tasks that have a postponed event inside the range.
- **Average duration** = mean of (finished_at - started_at) for tasks completed in the range that have both timestamps.
- **Streak** = consecutive days with at least one done task, ending today (or yesterday if today has none yet). Also show the best streak ever.

Summary cards: Completed (with the difference vs the previous period, for example +٣), Completion rate %, Postponed, Partially done, Waiting + Stopped, Average duration, Current streak and best streak.

Charts (fl_chart):

1. Bar chart of completed tasks: per day for a week, per week for a month, per month for a year. Tap a bar for a tooltip and to list those tasks.
2. Donut chart of the status distribution of the scheduled set, with the legend using status colors.
3. Top reasons: horizontal bars for the top 5 reasons across postponed / waiting / stopped / partial events in the range, with counts. Tapping a reason lists the tasks.
4. Most productive weekday: a small 7-bar chart of completed tasks by weekday.

Goals block: goals whose progress changed in the range (with the change in percentage points), milestones completed in the range, goals completed in the range.

Ideas block: new ideas in the range and ideas converted in the range.

Highlights list: 'أبرز ما اتنجز / Highlights' = the latest 10 completed tasks in the range (tap to open).

Empty state when the range has no data. Compute aggregates in SQL where possible so the screen stays fast with years of data.

8. Reminders and recurring tasks

Task reminders: optional per task: at the due time, 10 minutes, 30 minutes, 1 hour or 1 day before, or a custom time. If a task has a date but no time, a date-only reminder fires at 09:00 (configurable).

Daily summary (default on, 08:00): 'عندك N مهام ومشاوير النهارده / You have N items today'.

Evening review (default off, 21:00): 'راجع يومك وحدّث حالة مهامك / Review your day and update your tasks'. Tapping opens Today. This supports the owner's main goal: having honest feedback on what was done.

Notification action buttons: 'تمت / Done' and 'فتح / Open'.

Android details: request POST_NOTIFICATIONS on Android 13+ with a short explanation screen first. Use exact alarms only if the permission is granted, otherwise fall back to inexact alarms and tell the owner. Reschedule all reminders after device reboot, after app update, after restoring a backup, and after a timezone change.

Recurring tasks: in the task form, 'تكرار / Repeat': none, daily, weekly on chosen weekdays, monthly on a day of the month (clamped to the last day of short months), with an optional end date. The app generates real task instances for today and the next 7 days when it opens and once after midnight. Each instance is independent (its own status, history and reason). Editing a recurring task asks 'هذه فقط / This one only' or 'كل القادمة / All future'. Deleting the series stops future generation; past instances stay.

9. Backup, export and privacy

- All data lives only on the phone. If the app is uninstalled or the phone is lost, the data is lost, so backup is important. Show a clear warning in Settings > Backup.
- **Export backup**: a single JSON file with a schema version, containing every table. Share it through the Android share sheet or save to Files.
- **Import backup**: choose a file, show a preview with counts (tasks, goals, ideas...), then 'استبدال كل البيانات / Replace all data' after an explicit confirmation. Before replacing, automatically save a safety copy of the current data. v1 supports replace only (no merge).
- **CSV export**: tasks.csv (one row per task with all fields, UTF-8 with BOM so Arabic opens correctly in Excel) and events.csv.
- Optional weekly reminder to back up (default off).
- Privacy: no INTERNET permission in the release manifest; no analytics, no crash reporting services, no ads. Optional app lock (PIN or biometric through local_auth); when enabled, hide the app content in the recent-apps screen.

10. Coding rules for the agent

1. All UI strings live in ARB files (ar and en). No hardcoded text in widgets.
2. All colors, sizes and radii come from theme tokens (section 3). No hardcoded colors in widgets. Both themes must always work.
3. Every list and screen has loading, empty and error states with friendly Arabic and English messages.
4. Database changes only through Drift migrations. Never wipe or reset user data to fix a bug.

5. Business rules (status transitions, goal progress, dashboard numbers, recurrence generation, reminder scheduling) live in pure Dart services with unit tests. Write tests for every rule in sections 5, 7 and 8.
6. Format dates and numbers through one shared helper that respects the language and the numerals setting.
7. Accessibility: semantic labels on icon buttons, 48 dp targets, support font scale up to 1.3x, WCAG AA contrast in both themes.
8. Performance: lazy lists, indexed columns, aggregated SQL for stats, no heavy work on the UI thread.
9. Keep README.md (how to run on a phone, how to build the APK), CHANGELOG.md and DECISIONS.md (every decision and why) up to date.
10. Git: commit after every working feature with a clear message, and tag the end of each phase (phase-0, phase-1...). Never delete or rewrite working code without telling the owner why.
11. If the spec is silent or ambiguous, stop and ask the owner in Arabic. Offer 2 or 3 concrete options with a recommendation.

11. Build phases

After each phase: run the app on the owner's phone, give him a plain-Arabic test checklist, wait for his approval, commit and tag.

- **Phase 0 — Foundation**: project setup, theme tokens for light and dark, fonts, localization (ar and en) with RTL, full Drift schema with seed reasons, navigation shell with the 4 tabs, Settings with Language and Theme only. Done when: the app opens, language and theme switch live, layout mirrors in Arabic.
- **Phase 1 — Tasks core**: Today screen, quick add (task and errand), task card with checkbox, swipe and long-press, status sheet with every rule in section 5, task details with timeline and history, undo. Done when: a task can go todo > in_progress > partial > postponed > done with reasons, and the history shows every step.
- **Phase 2 — Organization**: Overdue, Inbox, Waiting and Postponed sections, History screen with search and filters, reasons manager, trash, multi-line paste import.
- **Phase 3 — Goals**: goals, milestones, numeric goals with progress log, linking tasks to goals and milestones, goal details.
- **Phase 4 — Ideas**: ideas, tags, filters, conversion to goal or task with links.
- **Phase 5 — Dashboard**: the full Stats screen from section 7 with unit-tested calculations.
- **Phase 6 — Reminders and recurring**: task reminders, daily summary, evening review, recurring tasks, reboot handling.

- **Phase 7 — Safety and polish**: backup, import and CSV export, app lock, trash purge, dark-theme QA, accessibility pass, performance pass, release APK.

12. Final acceptance checklist

1. Every item in the owner's original list works: task logging with start and finish times; partial, postponed, waiting and stopped states with reasons; long-term goals in parts; ideas; daily tasks and errands; weekly, monthly and yearly dashboard.
2. Marking a task done takes one tap. Postponing takes at most three taps including the reason.
3. History shows when each task was started, finished, postponed and why.
4. The dashboard numbers match a manual count on a test dataset of at least 50 tasks.
5. Arabic (RTL) and English (LTR) both look correct on every screen, in both light and dark themes.
6. The app works with airplane mode on and has no INTERNET permission.
7. A backup exported from the app can be imported back with identical data.
8. Reminders still fire after a phone reboot.
9. The app stays smooth with 5,000 tasks.
10. The README explains how to run and build the app in simple steps.

13. Assumptions to confirm (change any of them)

- App name is a placeholder: إنجاز / Enjaz.
- Week starts on Saturday. Numerals default to Western digits (123) with an option for Arabic-Indic (١٢٣).
- There is no 'cancelled' status: delete the task, or use 'stopped' with a reason.
- Postponing never moves the original due date silently: a postponed task keeps its due date and gets a postponed_until date.
- No cloud sync or accounts in v1. Android automatic cloud backup is turned off; the owner relies on manual backup files.
- A single user on a single device. Gregorian dates only.
- The app is installed as an APK directly. Publishing on Google Play is out of scope for v1.